

4.1 Analytical Techniques

Question Paper

Course	CIEA Level Chemistry
Section	4. Analysis
Topic	4.1 Analytical Techniques
Difficulty	Medium

Time allowed: 20

Score: /10

Percentage: /100

Question 1

Which alcohol is not likely to have a fragment ion at m/e at 43 in its mass spectrum?

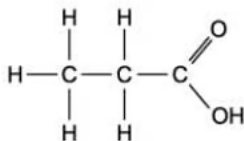
- A. $(\text{CH}_3)_2\text{CHCH}_2\text{OH}$
- B. $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CH}_3$
- C. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
- D. $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$

[1 mark]

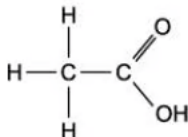
Question 2

Which of the compounds below is not likely to have a fragment ion at $m/e = 43$ in its mass spectrum?

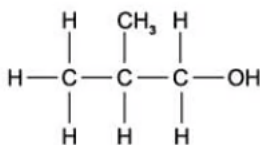
A.



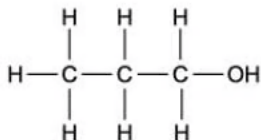
B.



C.



D.



[1 mark]

Question 3

Assuming that chlorine has only two isotopes and hydrogen and carbon as one, how many molecular ion peaks will be shown in the mass spectrum of $\text{C}_4\text{H}_6\text{Cl}_4$?

- A. 5
- B. 4
- C. 3
- D. 2

[1 mark]

Question 4

Bromine exists as two isotopes ^{79}Br and ^{81}Br , which are found in almost equal abundance.

Which of the following statements is correct?

- A. ^{79}Br is more reactive than ^{81}Br .
- B. The mass spectrum of $\text{C}_3\text{H}_7\text{Br}$ has two molecular ion peaks at 122 and 124.
- C. The atomic radius of ^{79}Br is less than the atomic radius of ^{81}Br .
- D. The first ionisation energy of ^{79}Br is less than the first ionisation energy of ^{81}Br .

[1 mark]

Question 5

Which alcohol is likely to have a fragment ion at $m/e = 31$ in its mass spectrum?

- A. $(\text{CH}_3)_2\text{CHCH}_2\text{OH}$
- B. $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CH}_3$
- C. $\text{CH}_3\text{CH}_2\text{CH}_2\text{C}(\text{OH})(\text{CH}_3)_2$
- D. $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$

[1 mark]

Question 6

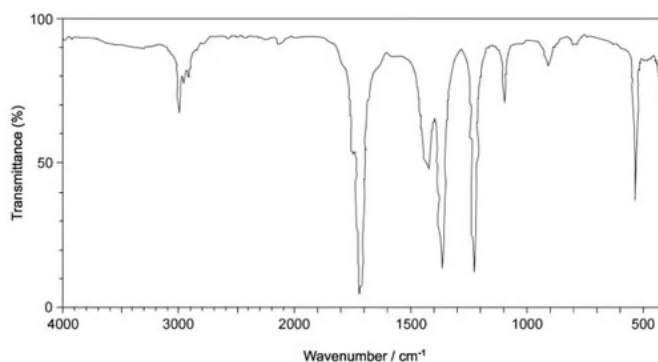
Which pair of compounds would you expect to both have a singly charged peak at $m/e = 29$ in the mass spectrum?

- A. Propan-1-ol and propanal
- B. Propanal and propanone
- C. Propan-2-ol and propanal
- D. Propan-1-ol and propan-2-ol

[1 mark]

Question 7

The infrared spectrum of a compound is shown below.



Use the infrared absorptions, in wavenumbers, to identify the compound.

Bond	Wavenumber range / cm^{-1}
O-H (alcohol)	3750 – 3200
C-H (alkane)	2962 – 2853
C-H (aldehyde)	2900 – 2820 and 2775 – 2700
C=O (aldehyde or ketone)	1740 – 1680

Which compound is shown by the infrared spectrum?

- A. Propan-1-ol
- B. Propan-2-ol
- C. Propanal
- D. Propanone

[1 mark]

Question 8

Which of the ketones listed would **not** be expected to have a peak in its mass spectrum at $m/e = 57$?

- A. Hexan-3-one, $\text{CH}_3\text{CH}_2\text{CH}_2\text{COCH}_2\text{CH}_3$
- B. Pentan-3-one, $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$
- C. 3-methylbutanone, $(\text{CH}_3)_2\text{CHCOCH}_3$
- D. Butanone, $\text{CH}_3\text{CH}_2\text{COCH}_3$

[1 mark]

Question 9

Chlorine has two isotopes, ^{35}Cl and ^{37}Cl .

What is the number of molecular ion peaks in the mass spectrum of a sample of Cl_2 ?

- A. 5
- B. 4
- C. 3
- D. 2

[1 mark]

Question 10

Which of the following statements about the mass spectrum of CH_3Br is correct?

- A. There is one peak for the molecular ion with an m/e value of 44.
- B. There is one peak for the molecular ion with an m/e value of 95.
- C. The last two peaks have abundances in the ratio 3:1 and occur at m/e values of 94 and 96.
- D. The last two peaks are of equal size and occur at m/e values of 94 and 96.

[1 mark]

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Model Answers: Medium

1

The correct answer is **D** because:

- A line at $m/e = 43$ corresponds to an ion with a mass of 43 for example:
 - $[\text{CH}_3\text{CH}_2\text{CH}_2]^+$
 - $[(\text{CH}_3)_2\text{CH}]^+$
- 2-butanol is not likely to have a fragment at $m/e = 43$ as it does not have a 3 carbon 7 hydrogen ion in its structure.

A is incorrect as the fragment $[(\text{CH}_3)_2\text{CH}]^+$ would produce an m/e peak at 43.

B is incorrect as the fragment $[\text{CH}_2\text{CH}_2\text{CH}_3]^+$ would produce an m/e peak at 43.

C is incorrect as the fragment $[\text{CH}_3\text{CH}_2\text{CH}_2]^+$ would produce an m/e peak at 43.

Exam tip: If not sure draw vertical lines in the compounds to isolate fragments.

2

The correct answer is **A** because:

- A line at $m/e = 43$ corresponds to an ion with a mass of 43 for example:
 - $[\text{CH}_3\text{CH}_2\text{CH}_2]^+$
 - $[\text{CH}_3\text{CO}]^+$
 - $[(\text{CH}_3)_2\text{CH}]^+$
- Propanoic acid is not likely to have a fragment at $m/e = 43$ as it does not have a 3 carbon 7 hydrogen ion in its structure.
- Propanoic acid will have a fragment at $m/e = 57$.

B is incorrect as this has the structure $[\text{CH}_3\text{CO}]^+$ with a mass of $m/e = 43$.

C is incorrect as this has the structure $[(\text{CH}_3)_2\text{CH}]^+$ with a mass of $m/e = 43$.

D is incorrect as this has the structure $[\text{CH}_3\text{CH}_2\text{CH}_2]^+$ with a mass of $m/e = 43$.

Exam tip: If you're not sure, draw vertical lines in the compounds to isolate fragments.

3

The correct answer is **A** because:

- The peak representing the **molecular ion (M^+ or $M_{\bullet}^+$)** is called the **molecular ion peak**. Excluding any peaks due to the presence of heavier isotopes, this is the peak with the highest m/e ratio.
 - This is the whole molecule just that has lost an electron.
- Chlorine has two isotopes, so there are 5 possible molecular ions:
 - $M - 4(^{35}\text{Cl})$
 - $M+2 - 3(^{35}\text{Cl}), 1(^{37}\text{Cl})$
 - $M+4 - 2(^{35}\text{Cl}), 2(^{37}\text{Cl})$
 - $M+6 - 1(^{35}\text{Cl}), 3(^{37}\text{Cl})$
 - $M+8 - 4(^{37}\text{Cl})$

4

The correct answer is **B** because:

- The peak representing the **molecular ion (M^+ or $M_{\bullet}^+$)** is called the **molecular ion peak**. Excluding any peaks due to the presence of heavier isotopes, this is the peak with the highest m/e ratio.
 - This is the whole molecule just that has lost an electron.
- Bromine has two isotopes, so there are 2 possible molecular ions:
 - $M - (^{79}\text{Br}) m/e = 122$
 - $M+2 - (^{81}\text{Br}) m/e = 124$

A is incorrect as isotopes' reactivities are almost identical.

C is incorrect as the differences in the number of neutrons do not affect atomic radii.

D is incorrect as the differences in the number of neutrons do not affect the ionisation energy.

5

The correct answer is **A** because:

D is incorrect as the differences in the number of neutrons do not affect the ionisation energy.

5

The correct answer is **A** because:

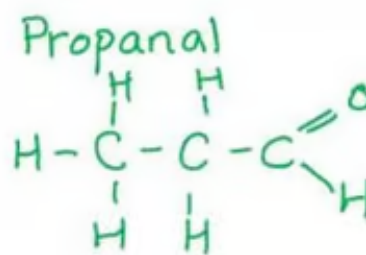
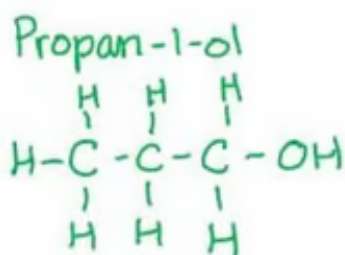
- A line at $m/e = 31$ corresponds to an ion with a mass of 31 for example:
 - $[\text{CH}_2\text{OH}]^+$
- 2-methyl-1-propanol is likely to have a fragment at $m/e = 31$ as it has a $[\text{CH}_2\text{OH}]^+$ ion.

Exam tip: If you're not sure, draw vertical lines in the compounds to isolate fragments.

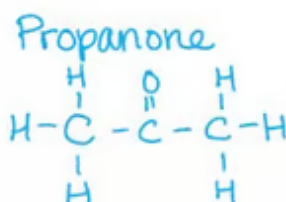
6

The correct answer is **A** because:

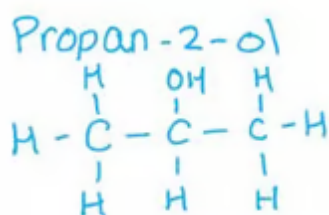
- A line at $m/e = 29$ corresponds to an ion with a mass of 29 for example:
 - $[\text{C}_2\text{H}_5]^+$
- The two compounds with this fragment are propan-1-ol and propanal.



B is incorrect as propanone does not have the $[\text{C}_2\text{H}_5]^+$ fragment.



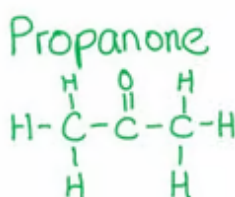
C & D are incorrect as propan-2-ol does not have the $[\text{C}_2\text{H}_5]^+$ fragment.



7

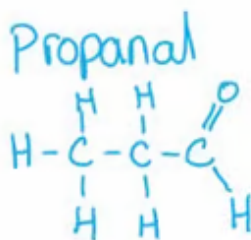
The correct answer is **D** because:

- Infrared is the analysis of infrared light interacting with a molecule; this can be measured by absorption, emission and reflection.
- It is used to determine functional groups in atoms based on the bond strength and mass of the atoms.
- The infrared spectrum shown in the question has a peak at 1680 – 1695.
- Propanone has a $\text{C}=\text{O}$ in its structure giving this characteristic peak.
- Comparing this to the data table in the question this matches the propanone structure.



A is incorrect as propan-1-ol would not have a peak at 1680 – 1695 and a broad peak at 3750 – 3200 corresponding to the alcohol group.

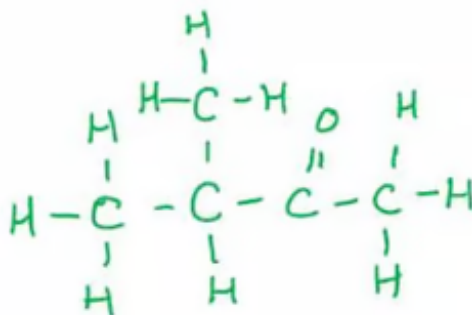
C is incorrect as propanal would have a peak at 1680 – 1695 but also a peak at 2900 – 2820 and 2775 – 2700 which is not present in this spectrum.



8

The correct answer is **C** because:

- A line at $m/e = 57$ corresponds to an ion with a mass of 57 for example:
 - $[C_4H_9]^+$
 - $[C_2H_5C=O]^+$
- 3-methylbutanone will not have either of these fragments.



A is incorrect as hexan-3-one has m/e at 57 due to the fragment ion $[C_2H_5C=O]^+$

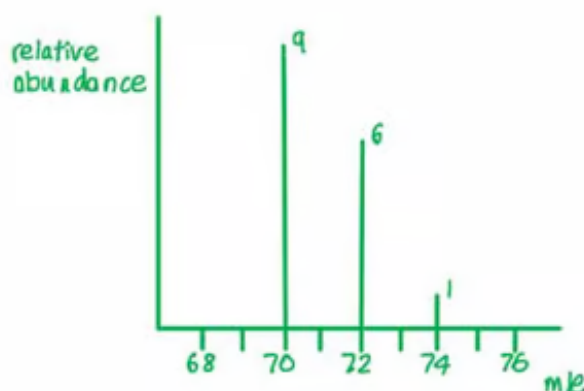
B is incorrect as pentan-3-one has m/e at 57 due to the fragment ion $[C_2H_5C=O]^+$

D is incorrect as butanone has m/e at 57 due to the fragment ion $[C_2H_5C=O]^+$

9

The correct answer is **C** because:

- Chlorine consists of molecules, not individual atoms. When chlorine is passed through the mass spectrometer, an electron is given off to give the **molecular ion, Cl_2^+** .
- Some of these will fragment to make Cl^+ + Cl^+
 - $\text{Cl}_2^+ \rightarrow \text{Cl}^+ + \text{Cl}^+$
- The Cl^+ atom passes through the machine and the Cl^+ ions will give lines at 35 and 37.
- There will also be a line for the unfragmented Cl_2^+ ion.
- This will give 3 molecular ion peaks
 - Cl_2^+ ion containing the isotopes $35 + 35 = 70$
 - Cl_2^+ containing the isotopes $35 + 37 = 72$
 - Cl_2^+ containing the isotopes $37 + 37 = 74$



A is incorrect as this is the total number of peaks including the fragments.

B is incorrect as this doesn't take into account the three possible molecular ions.

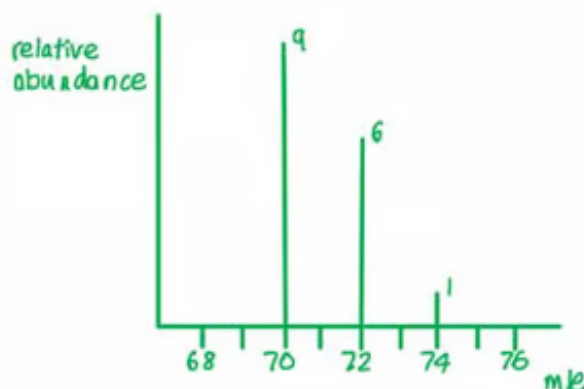
D is incorrect as this is the number of isotope peaks.

10

The correct answer is **D** because:

- Bromomethane will fragment to 3 peaks
 - $\text{CH}_3\text{Br} \rightarrow [\text{CH}_3^{81}\text{Br}]^+ + e^-$ at m/e 96

- Some of these will fragment to make $C/ + C^+$
 - $C_2^+ \rightarrow C/ + C^+$
- The $C/$ atom passes through the machine and the C^+ ions will give lines at 35 and 37.
- There will also be a line for the unfragmented C_2^+ ion.
- This will give 3 molecular ion peaks
 - C_2^+ ion containing the isotopes $35 + 35 = 70$
 - C_2^+ containing the isotopes $35 + 37 = 72$
 - C_2^+ containing the isotopes $37 + 37 = 74$



A is incorrect as this is the total number of peaks including the fragments.

B is incorrect as this doesn't take into account the three possible molecular ions.

D is incorrect as this is the number of isotope peaks.

10

The correct answer is **D** because:

- Bromomethane will fragment to 3 peaks
 - $CH_3Br \rightarrow [CH_3^{81}Br]^+ + e^-$ at m/e 96
 - $CH_3^{79}Br \rightarrow [CH_3^{79}Br]^+ + e^-$ at m/e 94
 - $[CH_3Br]^+ \rightarrow [CH_3]^+ + \bullet Br$ at m/e 15